Ai 4th Practical Code:-

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from collections import Counter
```

```
nltk.download('punkt')

nltk.download('stopwords')

nltk.download('averaged_perceptron_tagger')
```

```
text = "Artificial Intelligence and Machine Learning are important technologies in modern computer science."


print("Original Text:\n", text)
```

```
tokens = word_tokenize(text)

print("\nTokens:\n", tokens)
```

```
freq = Counter(tokens)

print("\nWord Frequency:\n", freq)
```

```
stop_words = set(stopwords.words('english'))

filtered_words = [word for word in tokens if word.lower() not in stop_words]

print("\nAfter Removing Stopwords:\n", filtered_words)
```

```
pos_tags = nltk.pos_tag(filtered_words)

print("\nPOS Tagging:\n", pos_tags)
```

Output:-

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Run Code

In [1]:
```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from collections import Counter
```

In [2]:
```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to C:\Users\Network
[nltk_data]     Lab\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package stopwords to C:\Users\Network
[nltk_data]     Lab\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\stopwords.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     C:\Users\Network Lab\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping taggers\averaged_perceptron_tagger.zip.
```

Out[2]: True

In [3]:
```
text = "Artificial Intelligence and Machine Learning are important technologies in modern computer scie

print("Original Text:\n", text)
```

```
Original Text:
 Artificial Intelligence and Machine Learning are important technologies in modern computer science.
```

In [4]:
```
tokens = word_tokenize(text)
print("\nTokens:\n", tokens)
```

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Run Code

In [4]:
```
tokens = word_tokenize(text)
print("\nTokens:\n", tokens)
```

```
Tokens:
 ['Artificial', 'Intelligence', 'and', 'Machine', 'Learning', 'are', 'important', 'technologies', 'in
', 'modern', 'computer', 'science', '.']
```

In [5]:
```
freq = Counter(tokens)
print("\nWord Frequency:\n", freq)
```

```
Word Frequency:
 Counter({'Artificial': 1, 'Intelligence': 1, 'and': 1, 'Machine': 1, 'Learning': 1, 'are': 1, 'impor
tant': 1, 'technologies': 1, 'in': 1, 'modern': 1, 'computer': 1, 'science': 1, '.': 1})
```

In [6]:
```
stop_words = set(stopwords.words('english'))
filtered_words = [word for word in tokens if word.lower() not in stop_words]
print("\nAfter Removing Stopwords:\n", filtered_words)
```

```
After Removing Stopwords:
 ['Artificial', 'Intelligence', 'Machine', 'Learning', 'important', 'technologies', 'modern', 'comput
er', 'science', '.']
```

In [7]:
```
pos_tags = nltk.pos_tag(filtered_words)
print("\nPOS Tagging:\n", pos_tags)
```

```
POS Tagging:
 [('Artificial', 'JJ'), ('Intelligence', 'NNP'), ('Machine', 'NNP'), ('Learning', 'NNP'), ('importan
t', 'JJ'), ('technologies', 'NNS'), ('modern', 'VBP'), ('computer', 'NN'), ('science', 'NN'), ('.',
'.')]
```